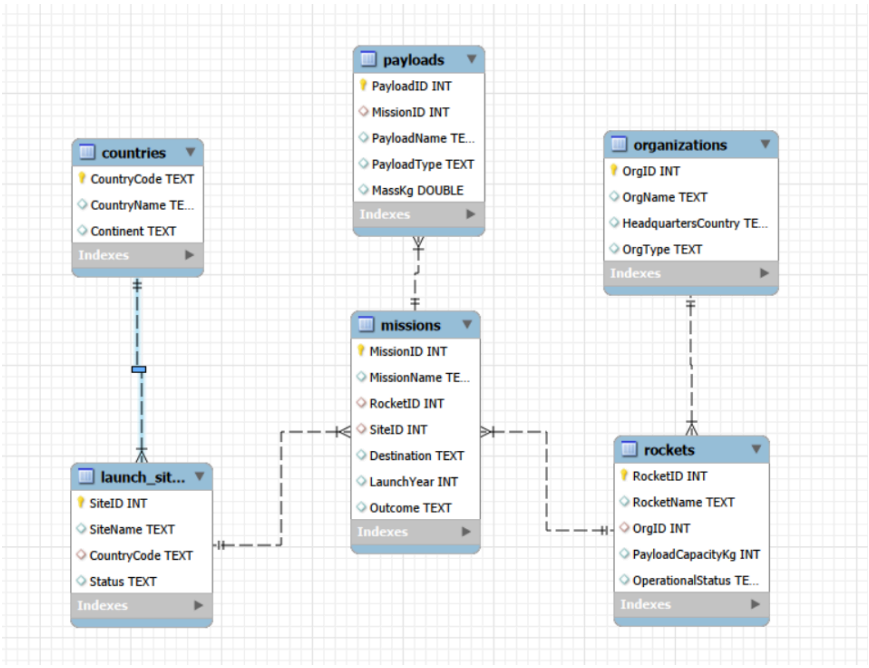
Uche Ogbodo
HW#4

EER Diagram:



Q1.

```sql
SELECT
    missions.MissionName,
    rockets.RocketName,
    organizations.OrgName,
    launch_sites.SiteName,
    missions.Destination,
    missions.LaunchYear,
    missions.Outcome
FROM missions
JOIN rockets
    ON missions.RocketID = rockets.RocketID
JOIN organizations
    ON rockets.OrgID = organizations.OrgID
JOIN launch_sites
    ON missions.SiteID = launch_sites.SiteID
ORDER BY
    missions.LaunchYear DESC,
    missions.MissionName ASC;
```

MissionName	RocketName	OrgName	SiteName	Destination	LaunchYear	Outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

Q2.

```sql
-- Q2
SELECT
    missions.MissionName,
    rockets.RocketName,
    payloads.PayloadName,
    payloads.PayloadType,
    payloads.MassKg
FROM payloads
JOIN missions
    ON payloads.MissionID = missions.MissionID
JOIN rockets
    ON missions.RocketID = rockets.RocketID
WHERE missions.LaunchYear = 2024
ORDER BY payloads.MassKg DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MissionName	RocketName	PayloadName	PayloadType	MassKg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100

Q3.

```sql
-- Q3
SELECT
    launch_sites.SiteName,
    missions.MissionName,
    rockets.RocketName,
    missions.LaunchYear
FROM launch_sites
LEFT JOIN missions
    ON launch_sites.SiteID = missions.SiteID
    AND missions.LaunchYear = 2024
LEFT JOIN rockets
    ON missions.RocketID = rockets.RocketID
ORDER BY launch_sites.SiteName ASC;
```

SiteName	MissionName	RocketName	LaunchYear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

Q4.

```sql
WITH Org_Payload_Summary AS (
    SELECT
        rockets.OrgID,
        SUM(payloads.MassKg) AS Total_Payload_Mass
    FROM payloads
    JOIN missions
        ON payloads.MissionID = missions.MissionID
    JOIN rockets
        ON missions.RocketID = rockets.RocketID
    GROUP BY rockets.OrgID
)
SELECT
    organizations.OrgName,
    organizations.HeadquartersCountry,
    organizations.OrgType,
    org_payload_summary.Total_Payload_Mass
FROM organizations
LEFT JOIN Org_Payload_Summary
    ON organizations.OrgID = Org_Payload_Summary.OrgID
ORDER BY Total_Payload_Mass DESC;
```

OrgName	HeadquartersCountry	OrgType	Total_Payload_Mass
National Aeronautics and Space Administration	USA	Government	25800
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200
Indian Space Research Organisation	IND	Government	150
Rocket Lab	NZL	Private	100
Arianespace	FRA	Private	NULL

Q5.

```sql
-- Q5
WITH mission_counts AS (
    SELECT
        launch_sites.countrycode,
        COUNT(missions.missionid) AS total_missions
    FROM launch_sites
    JOIN missions
        ON launch_sites.siteid = missions.siteid
    GROUP BY launch_sites.countrycode
),
payload_masses AS (
    SELECT
        launch_sites.countrycode,
        SUM(payloads.masskg) AS total_payload_mass
    FROM launch_sites
    JOIN missions
        ON launch_sites.siteid = missions.siteid
    JOIN payloads
        ON missions.missionid = payloads.missionid
    GROUP BY launch_sites.countrycode
)
SELECT
    countries.countryname,
    COALESCE(mission_counts.total_missions, 0) AS total_missions,
    COALESCE(payload_masses.total_payload_mass, 0) AS total_payload_mass
FROM countries
LEFT JOIN mission_counts
    ON countries.countrycode = mission_counts.countrycode
LEFT JOIN payload_masses
    ON countries.countrycode = payload_masses.countrycode
ORDER BY
    total_missions DESC,
    total_payload_mass DESC;
```

Result Grid | Filter Rows: | Export:

countryname	total_missions	total_payload_mass
United States	3	39492
Japan	1	6000
Russia	1	2450
China	1	1200
India	1	150
New Zealand	1	100
France	0	0
United Kingdom	0	0